When Models Surprise Us

Emergent Abilities, Chain-of-Thought, and the Thinking Revolution

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LLM Lab — Executive Education Program

2026

The Unifying Thread

"The dynamics of learning in LLMs violate our deepest intuitions."

A. Emergent Abilities

Scale a model 10× and it suddenly writes poetry.

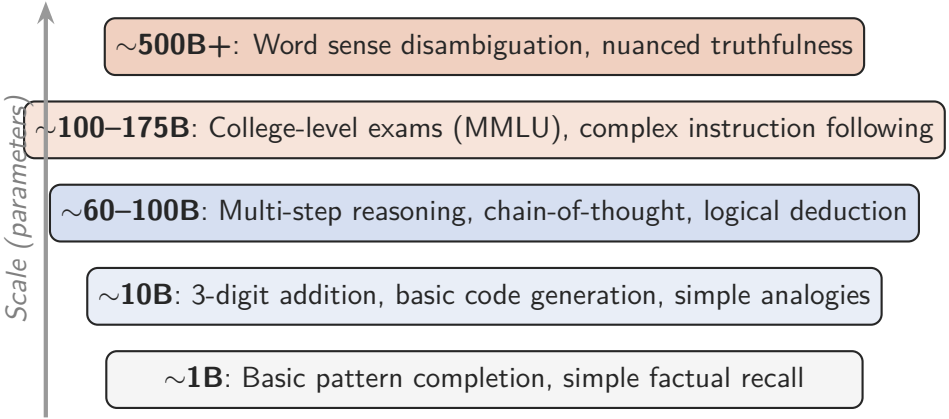
Or is that a measurement artifact?

B. Chain-of-Thought & Thinking Models

Ask a model to "think step by step" and it solves competition maths.

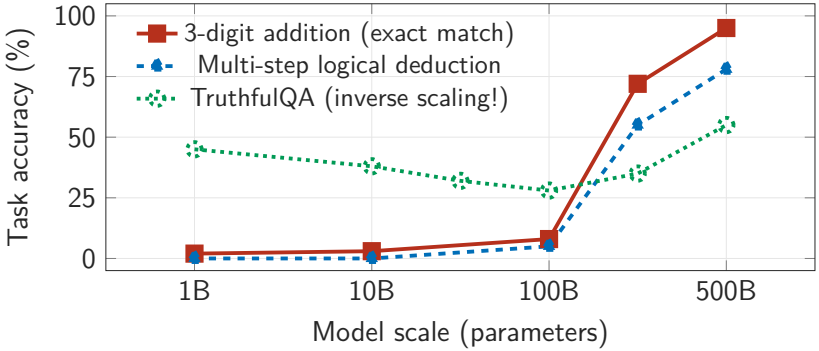
Train it with RL and it discovers reasoning on its own.

What Models Can Do at Each Scale



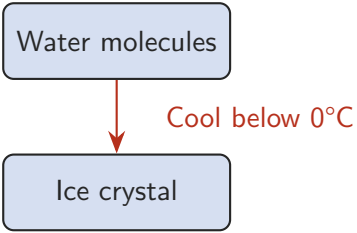
These are approximate thresholds from Wei et al. (TMLR 2022). Capabilities cluster at certain scales rather than appearing uniformly. [2]

Concrete Examples: From Nothing to Capability

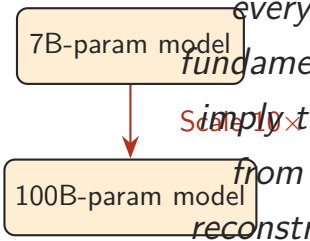


3-digit addition: 0% → 80%+ in a narrow range. **TruthfulQA**: medium models are less truthful than small ones — they've learned misconceptions. Only at 280B+ does truthfulness improve. [2, 3]

“More Is Different” — From Physics to AI



Same H_2O molecules, new macroscopic property



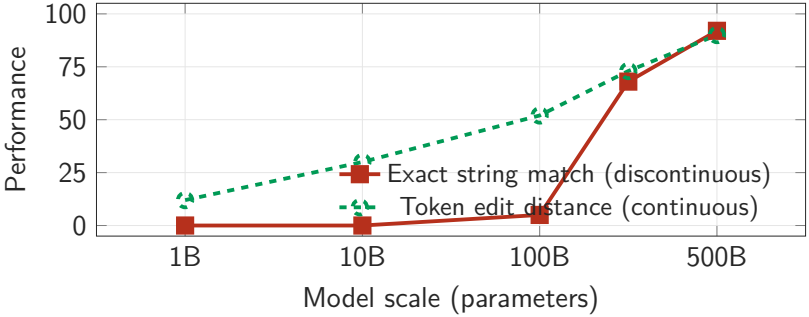
Same architecture, new cognitive capability

“The ability to reduce everything to simple fundamental laws does not imply the ability to start from those laws and reconstruct the universe.”

— P.W. Anderson, 1972 [1]

Nobel laureate. The same principle that explains why ice is rigid may explain why GPT-4 can write code

Or Is It a Mirage?



Schaeffer et al. (NeurIPS 2023, Outstanding Paper): 92% of claimed emergent abilities in BIG-Bench appeared only under **discontinuous metrics** (exact match, multiple-choice grade). Switch to a continuous metric $\Rightarrow$ smooth, predictable improvement.

Capabilities can appear suddenly at scale.

But what if you could unlock reasoning *without* scaling the model —

just by asking it to think step by step?

Chain-of-Thought: The Discovery That Changed Everything

Standard prompting:

```
Q: Roger has 5 tennis balls.
He buys 2 cans of 3 balls.
How many does he have?

A: 11
```

Chain-of-thought:

```
A: Roger started with 5.
2 cans × 3 = 6 new balls.
5 + 6 = 11.  The answer is 11.
```

Key results (PaLM 540B):

Method	GSM8K
Standard	∼18%
Chain-of-Thought	∼58%
Prior SOTA (fine-tuned + verifier)	55%

Zero fine-tuning. Just 8 hand-written exemplars with reasoning steps. [5]

Scaling Up: From One Trick to a Toolkit

Zero-shot CoT (Kojima et al., 2022)

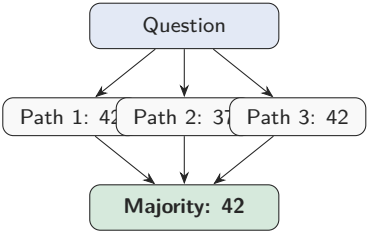
Just append:

"Let's think step by step."

GSM8K: 10.4% → **40.7%**
MultiArith: 17.7% → **78.7%**

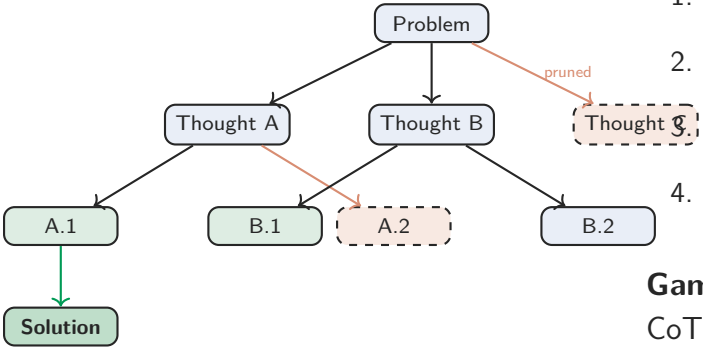
A single sentence — no exemplars — produces most of the benefit. [6]

Self-Consistency (Wang et al., 2023)



Sample N reasoning paths, majority vote on final answer.
GSM8K: **+17.9%** over CoT.
No extra training needed. [7]

Tree of Thoughts: Reasoning as Search



Yao et al. (NeurIPS 2023)

Organise reasoning as a **tree**:

1. Generate multiple thoughts
2. LLM *evaluates* each
3. BFS/DFS to explore
4. Backtrack when stuck

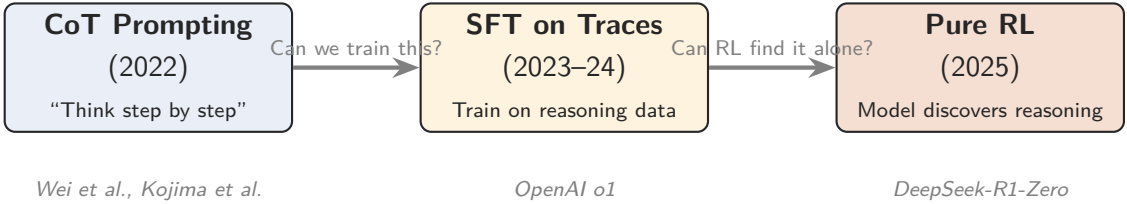
Game of 24:

CoT: **4%**

Tree of Thoughts: **74%**

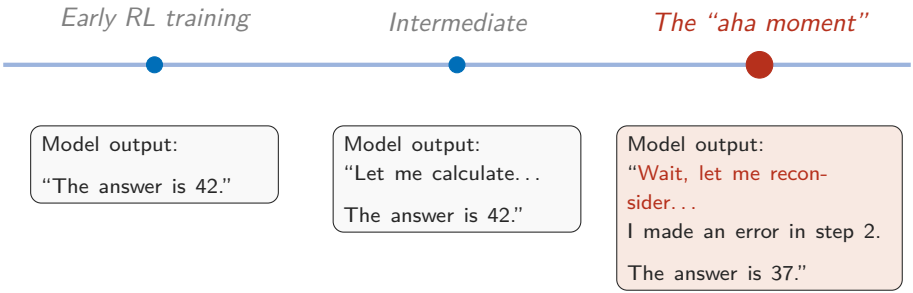
18.5× improvement from

The Leap: From Prompting to Thinking Models



The trajectory: a prompting trick (2022) → trained reasoning models (2024) → models that discover reasoning from scratch via RL (2025). This took less than 3 years.

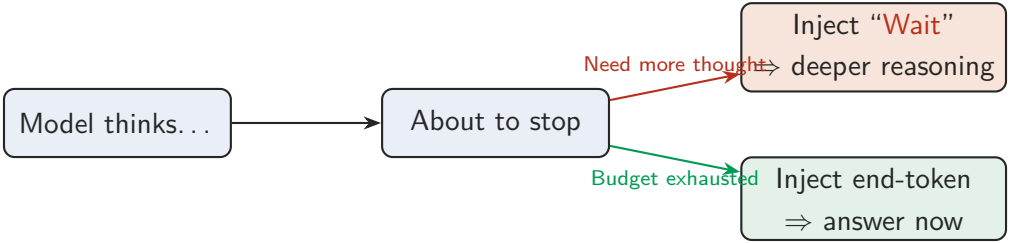
The "Aha Moment": Reasoning from Pure Trial and Error



DeepSeek-R1-Zero (2025): trained with *pure* reinforcement learning — no human demonstrations of reasoning. The model independently discovered that pausing to self-correct improves its answers. [9]

AIME 2024: 15.6% → **71.0%** through RL training alone. With majority voting: **86.7%** — matching OpenAI o1.

Budget Forcing: Controlling How Long a Model Thinks



s1-32B (Muennighoff et al., 2025):
Fine-tuned on only **1,000 examples**.
Training: **26 minutes** on 16 H100s. [10]

Model	AIME 2024
o1-preview	44.6%
s1-32B	50.0%
s1-32B + Wait×6	57.0%

The Test-Time Compute Landscape

Search-based

- Best-of-N
- Tree of Thoughts
- Self-Consistency
- Beam search

Refinement-based

- Self-correction
- Budget forcing
- Sequential revision
- "Wait" injection

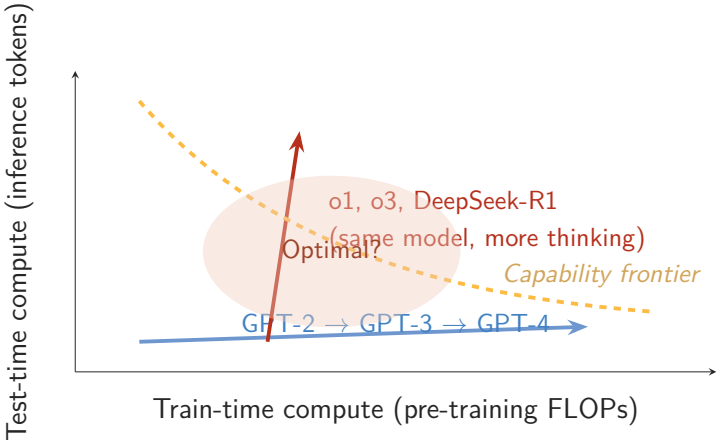
Verification-based

- Outcome RMs
- Process RMs
- Step-level scoring
- Generative PRMs

Hybrid / Learned: o1, o3, DeepSeek-R1, MCTS+PRM

Key insight (Snell et al., ICLR 2025): A small model with optimal test-time compute can outperform a 14× larger model. The optimal strategy is prompt-dependent. [11]

Two Axes of Scaling



The capability frontier now has *two dimensions*. The strategic question for any organisation deploying LLMs: where on this frontier should you invest?

Questions to Take Home

1. If we cannot predict what the next model will be capable of, how should organisations plan for AI adoption?
   (Emergent abilities mean capability forecasting is unreliable.)

2. Chain-of-thought evolved from a prompting trick to a training paradigm in under 3 years. What reasoning capabilities are next?
   (If pure RL can discover self-correction, what else can it discover?)

3. Test-time compute trades latency for accuracy. Which of your business decisions warrant a model that "thinks longer"?
   (Not all queries are equal — when is 30 seconds of thinking worth the cost?)

The dynamics of learning in LLMs violate our deepest intuitions.

Plan accordingly.

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